

Train, Validation, and Test Sets

1. Naive Forecasting (Baseline)

- **Method:** Predict the next value as the **same as the last observed value**.
 - **Purpose:** Simple baseline for comparing more complex models.
 - Surprisingly effective in some cases.
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2. Fixed Partitioning (Standard Approach)

- **Split** time series into:
 1. **Training period** → Model learns patterns.
 2. **Validation period** → Tune architecture & hyperparameters, measure performance.
 3. **Test period** → Final unbiased performance check.
 - **Important:**
 1. If seasonality exists, ensure **whole seasons** in each split (e.g., full years).
 2. Avoid random shuffling (unlike non-time-series datasets).
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3. Training Workflow

1. Train on **training set** → Evaluate on **validation set**.
 2. After tuning, **retrain on training + validation data** → Test on **test set**.
 3. If needed, **retrain again including test data** → Helps since it's closest to the present and thus more relevant for predicting the future.
 4. In some cases, skip the test set entirely and treat **future data** as the “real” test.
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4. Roll Forward Partitioning (Alternative)

- Start with **short training period**, expand gradually (e.g., +1 day/week each iteration).
 - At each step:
 1. Train on training period.
 2. Forecast next time step in validation.
 - Like repeated fixed partitioning, **continuously refining** the model.
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➤ Key Insight:

For time series, **data order matters**. You can't shuffle. Whether you use fixed or roll forward partitioning, keep the chronological sequence intact.
